Domain-Specific Financial LLMs

Lecture 08 — When Finance Needs Its Own Models

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Outline

Learning Objectives

By the end of this lecture you should be able to:

1. Explain *why* general-purpose LLMs underperform on financial text, and name the linguistic features that create the gap.
2. Place any financial model in a taxonomy: encoder-only vs. decoder-only, from-scratch vs. domain-adaptive vs. fine-tune-only.
3. Apply **domain-adaptive pre-training (DAPT)** and design a balanced corpus.
4. Read FinQA / FLUE / FinBen results in terms of the task taxonomy.
5. Weigh licensing, compute, governance, and accuracy when choosing a model for production.

The Central Tension

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General models have *breadth* — hundreds of billions of web tokens give them world knowledge about options, CDS, and Basel III. Finance demands *specificity* — a model that confuses “basis risk” with “basis points”, misreads an income-statement table, or reads “liability” with its everyday connotation produces outputs that are not merely imprecise but **dangerous** downstream.

The resolution is not breadth *vs.* depth but *how to combine them*. Domain-specific financial LLMs orient pre-training, fine-tuning, or both toward the linguistic distribution of financial text.

First question: what actually makes financial language different?

Financial Language: A Distinct Register

Three structural features underrepresented in general corpora

1. **Lexical specificity.** Common words carry precise, sometimes *reversed* meanings: "liability", "short", "haircut". The Loughran–McDonald dictionary catalogues ∼350 words whose polarity is neutral or inverted in SEC filings vs. general English.
2. **Numeric density.** "Revenue grew 12.3% YoY to $4.7B" is fluent English *and* an arithmetic claim. FinQA requires multi-step reasoning over tables from 10-Ks.
3. **Tabular embedding.** Income statements, balance sheets, loan tapes encode meaning in 2-D grids; standard tokenisation flattens them and loses the row/column alignment.

A model is *financially literate* to the extent it resolves token meaning on all three dimensions.

Vocabulary Mismatch is a Tokenisation Problem

UNDER THE HOOD

The mismatch operates at the **sub-word** level. A BPE tokeniser trained on general text may split `EBITDA` into 4–5 sub-tokens, none of which carries financial meaning — the burden shifts onto the attention layers to reassemble it.

- Tokenisers trained on financial corpora represent abbreviations (`EBITDA`, `CUSIP`, `LIBOR`) as single tokens.
- BloombergGPT trained a *custom* BPE tokeniser on its combined general+financial corpus precisely so domain tokens stay compact.

Why it matters

Fewer tokens per financial term $\Rightarrow$ shorter sequences, lower compute, and crucially a single embedding the model can specialise — instead of a fragile composition over fragments.

How General Models Fail — Four Concrete Modes

Failure mode	**Evidence**
Sentiment polarity inversion	"strong", "leverage", "unprecedented" flip valence; general model fine-tuned on Financial PhraseBank trails FinBERT despite *more* parameters.
Numerical hallucination	GPT-class models fabricate EPS / figures that diverge from the source 10-K (Kang & Liu, 2023).
Table-parsing failure	On FinQA, GPT-3 (175B) zero-shot $<60\%$; fine-tuned specialists $>80\%$. Prompting alone does not close the gap.
Regulatory ambiguity	ISDA / Basel / SEC comment letters are a legal sub-dialect; SEC-BERT beats general BERT on NER and clause classification.

Common root

All four reflect the gap between the financial distribution and the web-text distribution that dominates pre-training. The remedy: shift the model's prior

Two Axes of Classification

THE BIGGER PICTURE

Every financial model is a point in a 2-D design space. Where you land dictates compute cost, capability profile, and deployment constraints.

Axis 1 — Architecture

- Encoder-only (bidirectional): classification, extraction
- Decoder-only (autoregressive): generation, multi-step reasoning
- Encoder–decoder (seq2seq): summarisation, table-to-text

Axis 2 — Training strategy

- From scratch (BloombergGPT)
- Domain-adaptive continuation / DAPT (FinBERT)
- Fine-tune / instruction-tune only (FinMA, InvestLM)

Definition (Financial LLM)

A model is *financial* if its pre-training corpus has a significant financial fraction, *or* it is fine-tuned on financial NLP tasks, *or* its tokeniser is

Encoder–Decoder: the T5 Family in Finance

UNDER THE HOOD

A bidirectional **encoder** reads the full source; an autoregressive **decoder** emits output token by token. The separate encoding pass lets the model attend to a long source *before* writing a single output token — reducing copy errors and improving factual consistency.

Canonical model: **T5** / instruction-tuned **FLAN-T5**. Financial uses:

- Earnings-call summarisation (hours of commentary → structured abstract)
- Abstractive 10-K summarisation (100 pages → one-page narrative)
- Financial table-to-text generation

When still preferred: constrained-output, extraction-heavy summarisation, even though large decoder-only models (FinMA) now also handle these at scale.

Encoder Family: FinBERT, SEC-BERT, FinBERT-tone

Model	Base / corpus	Specialisation
FinBERT (Araci 2019)	BERT-base, ∼1.8B words: Reuters TRC2, Bloomberg news, PhraseBank	3-class sentiment; large F1 gain over BERT-base on PhraseBank
FinBERT-tone (Yang et al. 2020)	BERT-large, earnings calls + analyst reports + filings	LM tone: optimistic / pessimistic / litigious / uncertain / constraining
SEC-BERT (Loukas 2022)	BERT-base, ∼3B tokens of EDGAR only (10-K/Q, 8-K, proxies)	NER + clause classification on regulatory text

Why default to encoders for classification?

110–340M params, open weights, well-understood fine-tuning. EDGAR is a coherent legal sub-dialect — a news-trained model does *not* fully transfer to it.

Worked Example: FinBERT-tone on an Earnings Call

Transcript excerpt

"While we are **cautiously optimistic** about the trajectory of our core business, we want to flag meaningful **headwinds** in the European segment, driven by continued **softness** in consumer demand and heightened **competitive pressure**."

General sentiment model
Labels **positive** — "optimistic" is a strong positive signal in product-review / social-media text.

FinBERT-tone
Labels **uncertain / pessimistic** — "cautiously optimistic" is a hedge; "headwinds", "softness" are near-universally negative in this register.

Downstream stakes

This tone feeds analyst sentiment indices that drive systematic trading strategies (Tetlock 2007). A label error here is a position error.

Decoder Family: BloombergGPT, FinGPT, FinMA, InvestLM

Model	**Scale**	**Approach**
BloombergGPT	50B params, 708B tokens	From scratch; ∼50/50 finance (363B FinPile) / general (345B Pile, Wiki, books). Beats peers on financial NLP, matches GPT-NeoX-20B on general tasks.
FinGPT	LoRA on LLaMA/ChatGLM/Falcon	Open-source, *data-centric*; fine-tune on a single A100 with live news APIs. Democratisation over peak accuracy.
FinMA (PIXIU)	7B LLaMA	Instruction-tuned on 136k financial instruction–response pairs across 5 task types. Strong in-distribution; weaker on unseen formats.
InvestLM	65B LLaMA	Instruction-tuned on a *small, curated* set (CFA Qs, research reports). Large base beats 7B models with fewer examples.

Decoder Family — the Autoregressive Objective

UNDER THE HOOD

Decoder LLMs are autoregressive: $\mathcal{L}_{\text{CLM}} = -\sum_t \log p_\theta(w_t \mid w_{<t})$. The roughly 50/50 corpus split in BloombergGPT is deliberate — a finance-only model loses the world knowledge that makes an LLM useful on finance tasks requiring general reasoning.

Reading the four models as a design spectrum

- **BloombergGPT** buys peak accuracy with from-scratch compute few can match.
- **FinGPT** trades peak accuracy for reproducibility on a single A100.
- **FinMA** is strong in-distribution but brittle on unseen formats.
- **InvestLM** shows a large base + small curated set can beat 7B models.

Encoder vs. Decoder — a Practical Heuristic

Decision rule

- **Fixed label space** (pos/neg/neutral, default/no-default)? ⇒ start with an **encoder** (FinBERT, SEC-BERT): compact, fast, calibrated.
- **Open-ended output** (draft a risk section, answer from a 10-K)? ⇒ a **decoder** is necessary.

- Encoders: 110–340M, fast to fine-tune, well-understood.
- Decoders: an order of magnitude larger, more inference memory, harder to calibrate for structured prediction.
- **Hybrid:** a small decoder + a well-indexed document store (RAG) can substitute for a much larger model (Lecture on LLM agents).

From Scratch or Continue? The Build Decision

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When Bloomberg built BloombergGPT it faced the choice every team confronts: train a new model on a mixed corpus, or start from a general model and *continue* pre-training on financial text?

From scratch

- Clean control of tokeniser, positional encoding, corpus weights
- Requires compute + data that *very few* organisations possess

Continue (DAPT)

- Reuses learned representations of a general foundation model
- Extends toward the domain at a fraction of the cost
- The dominant strategy in practice

Domain-Adaptive Pre-training (DAPT)

UNDER THE HOOD

Definition. Given pre-trained parameters θ_0 of model M_0 and an unlabelled domain corpus $\mathcal{D}_{\text{domain}}$,

$$\theta^{\star} = \arg\min_{\theta} \mathcal{L}_{\text{PT}}(\theta; \mathcal{D}_{\text{domain}}),$$

where $\mathcal{L}_{\text{PT}}$ is MLM (encoders) or CLM (decoders), the optimisation is *initialised at* θ_0, and the learning rate is reduced ($\sim 10\times$) to mitigate catastrophic forgetting. Then fine-tune $M^{\star}$ on labelled task data.

- Gururangan et al. (2020): consistent downstream gains across 4 domains; *largest* when the domain is most dissimilar from pre-training.
- Finance is strongly out-of-distribution for web-text models $\Rightarrow$ DAPT gains should be large.
- Budget: typically **1–10B** domain tokens vs. hundreds of billions for original pre-training.

DAPT: How Much Data, and Two Refinements

Diminishing returns. The largest gain is in the first ∼1B domain tokens; smaller increments thereafter.

- Encoders (110–340M): **1–5B** domain tokens usually suffice.
- Decoders (7–70B): need proportionally larger corpora.

TAPT (task-adaptive)

Further pre-train on *unlabelled* examples from the task distribution before supervised fine-tuning. Consistent gains, but needs task-specific unlabelled data.

Catastrophic forgetting

Narrow-domain continuation can degrade general ability. Mitigate with a reduced LR, a large enough corpus, and **replay** — mixing a small fraction of general data back in.

Hirano (2024): Japanese financial LLMs confirm DAPT gains generalise across languages and sub-domains.

Corpus Composition: Sources for Financial Pre-training

Source	Scale	Quality	Access	Coverage
Bloomberg News	∼300B	Very high	Proprietary	News, markets
SEC EDGAR	∼50B	High	Open	Regulatory, corporate
Reuters TRC2	∼2B	High	Licensed	Financial news
Financial PhraseBank	∼5M	High	Open	Sentiment-labelled
Seeking Alpha	∼10B	Medium	Open/API	Retail analysis
Earnings transcripts	∼5B	High	Commercial	Corporate comms
SSRN / arXiv finance	∼2B	High	Open	Academic research
General web (Pile)	∼800B	Mixed	Open	Broad coverage

The defining fact

General web text (∼800B) dwarfs even the largest *financial* source. This asymmetry is exactly *why DAPT beats from-scratch*: cheaply shift a model that already has the breadth, rather than rebuild it.

Example: A Central-Bank Communication Model

Goal: classify FOMC paragraphs as hawkish vs. dovish — needs the precise vocabulary of monetary policy ("accommodation", "balance-sheet normalisation", "neutral rate").

A deliberately narrow corpus

1. FOMC statements + minutes 1994–present ($\sim$50M tokens) — target distribution.
2. Other central-bank publications: ECB, BoE, Beige Books ($\sim$200M).
3. Monetary-policy academic papers, SSRN/NBER ($\sim$500M).
4. Financial news mentioning policy keywords (1–2B).

This model would lose on general financial tasks — but on monetary-policy communication it *beats* a general model fine-tuned on the same labels, because pre-training aligns with both the vocabulary and the pragmatics of central-bank language.

The Standard Financial NLP Benchmarks

Bench	What it tests	Headline numbers
FinQA	8,281 QA pairs over 10-K/Q tables + prose; multi-step arithmetic. The hardest standard benchmark.	Fine-tuned specialists ∼68–72% exec. acc.; zero-shot GPT-4 class ∼55–60%.
FLUE	5-task NLU suite (GLUE analogue): sentiment, headline class., NER, FiQA QA, STS.	FinBERT > general BERT/RoBERTa; biggest margins on sentiment + NER.
FinBen	36 datasets, 24 tasks, 5 capability categories incl. generation + risk mgmt.	Designed to test *decoder* open-ended generation, not just classification.

How to read the headline numbers

Each benchmark stresses a different capability. FinQA isolates table + arithmetic reasoning; FLUE isolates lexical NLU; FinBen pushes open-ended generation. A model strong on one is *not* automatically strong on the others.

FinQA Scoring: Execution Accuracy

UNDER THE HOOD

Execution accuracy (FinQA):

$$\mathrm{EA} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbf{1}\big[\, |\hat{a}_q - a_q| \leq \epsilon \,\big],$$

$\epsilon = 0$ for exact match, or a small relative threshold ($\sim$1%) for rounding.

Why a tolerance at all?

A model can execute the right reasoning program yet differ from the gold answer by a rounding step (e.g. reporting a ratio to two decimals). A small relative ϵ credits correct *reasoning* without rewarding spurious over-precision.

When Do FinLLMs Win? When Do They Lose?

FinLLMs win

Tasks tied to domain *lexical semantics*:

- PhraseBank sentiment — FinBERT $\gg$ BERT/RoBERTa, even after equal fine-tuning.
- EDGAR NER — SEC-BERT $\gg$ general BERT (entity surface forms differ).

General models win / tie

Tasks needing broad knowledge or deep reasoning:

- Macro / regulatory-history / geopolitical questions.
- FinQA with chain-of-thought on a large model often matches a smaller specialist — reasoning is a function of *scale*.

The Scaling Crossover

Domain advantage vs. model scale

The domain advantage is roughly *inversely proportional to model scale* at fixed difficulty.

- $\leq$7B: DAPT gives large, consistent gains.
- $\geq$70B: gains persist only for lexically sensitive tasks (sentiment, tone, NER); they fade for long-document reasoning.

A countervailing result

Rahimikia & Drinkall: year-specific FinLLMs can still beat far larger general models on return prediction — once one controls for look-ahead bias.

Two Caveats Every Benchmark Number Carries

Data contamination

A benchmark may already sit in a model's pre-training corpus. Several large models are *suspected* to have seen PhraseBank / FinQA train splits, inflating apparent performance. Rigorous evaluation needs held-out test sets with **temporal cutoffs** after the model's knowledge cutoff — which most published benchmarks do not enforce.

Hallucination is not solved by domain pre-training

It can be *worsened*: a model trained on financial news learns to generate plausible-but-unverified statistics. Kang & Liu (2023) find finance-trained models *more* likely to fabricate specific figures. Retrieval augmentation, constrained generation, and factual verification matter just as much for FinLLMs as for general models.

Deployment is Orthogonal to Benchmark Accuracy

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A model that tops FinQA but cannot be *licensed* commercially, needs a compute budget you cannot sustain, or produces outputs you cannot *trace* to source for audit — is not a deployable model.

Three consequential dimensions:

1. Licensing & data governance
2. Hosting strategy (cloud API vs. on-prem)
3. Cost-vs-performance $\Rightarrow$ hybrid architectures

Licensing, Hosting, Governance

Model	License
BloombergGPT	Proprietary, not released (terminal subscriber
FinBERT (Araci); FinBERT (Yang)	Apache 2.0 — commercial use OK
FinGPT	MIT
InvestLM / FinMA	Research-only (LLaMA-1 / LLaMA-2 license

Data governance

GDPR / GLBA / CCPA constrain training on customer data. Customer balances, transactions, credit scores generally *cannot* train a model for *other* customers without consent. **Synthetic data** is a common workaround — at the cost of distribution shift.

Hosting. FinBERT (110–340M): single A10G, sub-50ms, fits in memory. 7–50B decoders: multi-GPU; **quantisation** (GPTQ, AWQ, QLoRA) cuts memory 2–4× to fit 13–30B on one server. 50B+ : cluster-scale, few institutions.

Cost vs. Performance: the Effective-Cost Model

UNDER THE HOOD

Effective cost per correct output. With per-query model cost c_m, human review cost c_h, model correctness p_m, and routed fraction ρ:

$$C_{\text{eff}}(m, \rho) = \frac{c_m + \rho\, c_h}{p_m + \rho(1 - p_m)},$$

assuming human review recovers all errors. Minimise subject to $p_m + \rho(1 - p_m) \geq p^\star$.

Hybrid routing pattern

A small, fast FinLLM handles the easy majority; low-confidence queries route to a larger model or a human. Effective when difficulty is **skewed** — a large "easy" head + a small "hard" tail.

Cost vs. Performance: RAG as a Complement

Grounding instead of growing

A compact 7–13B decoder grounded in a well-indexed EDGAR store can match a larger ungrounded model on document tasks — at lower cost and with **full traceability** (an SR 11-7 / regulatory requirement).

- The retrieved passage pins generated figures to source text, cutting numerical hallucination.
- Traceability is not a nicety: model-risk regimes *require* an auditable link from output back to evidence.
- This makes a small grounded model a regulatory-grade alternative to a large ungrounded one.

Example: Cost-Optimal Regulatory-Filing Classifier

Task: classify "risk factor" sections of 10-Ks into 15 categories for 2,000 companies/year; 3–10 paragraphs each (≈20,000–60,000 paragraphs).

Option	Accuracy	Annual co
70B general model via API (all paragraphs)	93%	\$200–\$600
Fine-tuned FinBERT-base on a T4 GPU	88%	∼\$400 ser
Hybrid (FinBERT + route 12% low-conf. to 70B)	**92%+**	∼**\$250**

Lesson

The hybrid dominates either model *alone* on the accuracy–cost frontier. The break-even point is highly sensitive to the *confidence distribution*, which must be measured on a held-out validation set — not assumed.

Lecture 08 — Key Takeaways

1. **Register.** Financial language = specialised vocabulary + numeric density + tabular structure ⇒ a systematic general-model gap. The LM dictionary quantifies the inversion starkly.
2. **Two families.** Encoders (FinBERT, SEC-BERT) for fixed-label classification; decoders (BloombergGPT, FinGPT, FinMA, InvestLM) for open-ended generation and reasoning.
3. **DAPT** is the broadly applicable strategy: continue the PT objective on a financial corpus blended with enough general text.
4. **Benchmarks.** FinLLMs win on lexically sensitive tasks; lose the edge at large scale on reasoning. Watch contamination and hallucination.
5. **Deployment.** License audit, governance, hosting, and the accuracy–cost trade-off. **Hybrid routing** dominates single-model deployments.

Appendix

Extended Material: Architectures, Sources, Mitigations

Extra / optional material — beyond the core lecture.

Appendix A — Encoder–Decoder Internals

Full encoder–decoder (T5) processing of a financial document:

1. **Bidirectional encoder** attends over the entire 10-K source $\Rightarrow$ each source token sees full context.
2. **Cross-attention** lets each decoder step query the encoded source.
3. **Autoregressive decoder** emits the summary token by token.

Why factual consistency improves

Because the encoder consumes the *whole* source before generation starts, the decoder can copy figures and entities against a complete representation — fewer fabricated numbers in extraction-heavy summarisation than a pure decoder reading left-to-right.

Appendix B — Replay & Forgetting Detail

Catastrophic forgetting: continued pre-training on a narrow corpus degrades general language ability.

Three mitigations, increasing cost:

1. **Reduced learning rate** ($\sim 10\times$ below original PT): cheapest; limits how far weights drift.
2. **Sufficiently large domain corpus** relative to total tokens processed: keeps the update well-conditioned.
3. **Replay**: interleave a small fraction of original general-corpus data with the domain corpus — modest extra cost, materially less forgetting.

TAPT can be stacked on top: DAPT (broad domain) $\rightarrow$ TAPT (task-distribution, unlabelled) $\rightarrow$ supervised fine-tuning.

Appendix C — Secondary Corpus Notes

- **Bloomberg archive:** decades of professionally edited news — consistent terminology, calibrated tone, high factual density. The gold standard, but proprietary (barrier for academia / open source).
- **SEC EDGAR:** free since 1996; full US public-company universe; continuously updated. Caveat: uneven quality (small-firm boilerplate) and XML structure needs careful parsing to split prose from tables/metadata.
- **News aggregators** (Reuters, AP, Seeking Alpha, subreddits): broader topical coverage, lower editorial quality. Reuters TRC2 is a standard source.
- **Analyst reports:** among the richest, most precise texts — almost entirely proprietary; licensing at pre-training scale is cost-prohibitive.

Appendix D — Quantisation for On-Prem Decoders

To fit a 13–30B decoder on a single high-memory GPU server:

Method	**Idea**	**Effect**
GPTQ	Post-training weight quantisation (4-bit)	2–4$\times$ memory reduction
AWQ	Activation-aware weight quantisation	Preserves salient channels
QLoRA	4-bit base + LoRA adapters in fine-tuning	Train + serve cheaply

All trade a modest accuracy cost for a large memory saving — the enabling technology for on-prem deployment of mid-scale decoders where data residency forbids a cloud API.

Appendix E — Hallucination Mitigation Stack

Domain pre-training does *not* remove hallucination. Production stack:

1. **RAG**: ground generation in retrieved, verified passages.
2. **Constrained / structured generation**: schema-enforced outputs.
3. **Self-consistency**: K samples, majority vote (numerical reasoning).
4. **Claim decomposition** + automated factual verification against source.
5. **Confidence elicitation + abstention**: route low-confidence to humans.

Governance hook

SR 11-7 model-risk management and (forthcoming lecture) the EU AI Act / MiFID II make *traceability* a requirement — which is itself an argument for RAG-grounded architectures over ungrounded large models.